

Raghav Verma

New Delhi, India | raghav.verma.work@gmail.com | GitHub | raghav-verma.com | LinkedIn

Technical Skills

Languages	TypeScript, JavaScript (ES6+), Python, SQL
Frontend	React 18, Next.js 14, Vite, Tailwind CSS, shadcn/ui, HTML5, CSS3
CV / AI	YOLOv8, OpenCV, pose estimation, object tracking, temporal motion analysis, geometric feature extraction
Backend	Node.js, Express, Socket.IO, WebSocket, REST APIs, PostgreSQL, Prisma ORM
Systems	FFmpeg, Docker, Python CLI, shell scripting, Git, Electron
Visualization	Three.js, React Three Fiber, WebGL, Canvas API
Testing	Postman, Jest, Vitest

Professional Experience

SDE Intern — **Khel.AI** Feb 2026 – Present

- Architected a low-latency edge ingestion system using Express, OBS WebSocket integration, and Socket.IO pipelines for real-time broadcast processing.
- Built FFmpeg-based video processing workflows with transcoding, stream-copy clipping, atomic writes, and automated cache cleanup.
- Developed an end-to-end cricket video analytics pipeline using YOLOv8, OpenCV, and temporal motion analysis to automatically detect and clip bowling deliveries from full-match footage.
- Engineered multi-stage delivery detection using wrist apex tracking, foot-contact analysis, ball trajectory validation, replay suppression, and scene segmentation for noisy broadcast video.
- Built frame-level analytics and React review tooling for delivery classification, release tracking, no-ball estimation, metadata generation, and custom cricket ball detector training workflows.

Frontend Engineer (Contract) — **Hypeliv Solutions Pvt. Ltd** Aug 2025 – Jan 2026

- Improved Next.js performance reducing LCP from 2.5s to 1.5s via code splitting and lazy loading.
- Built real-time trading interfaces integrating WebSocket market streams.
- Defined type-safe API contracts with backend teams.
- Improved reliability of UI data handling through schema validation.

Frontend Engineer Intern — **The TechnoLabs** Jan 2024 – Jul 2024

- Built React invoice system with validation and PDF export.
- Integrated REST APIs for full invoice lifecycle.
- Automated OCR dataset validation using Python scripts.
- Improved data consistency checks using structured validation logic.

Projects

Forge — **Autonomous Operator Runtime** Founder

- Built a TypeScript monorepo platform for specialized AI operators (HR, Finance, Marketing, Support, Data, Dev) executing business workflows with mandatory approval gates and append-only audit logging.
- Architected permission-enforced operator registry, event bus coordination, persistent memory store, and 26 cross-operator routines (cron + event-triggered) on a 60s tick interval.
- Engineered approval-flow engine with proposal staging, pre-apply verification (tsc + ESLint), and impact scoring via dependency graph analysis for code synthesis workflows.
- Designed multi-client support (VS Code extension, web console, CLI) with TypeScript monorepo (strict dependency constraints), real integrations (Gmail, Slack, Linear, Sheets, Stripe, GitHub), and session replay with cost telemetry.

Open Source Projects

Meridian — **API Resilience SDK** npm | GitHub

- TypeScript SDK providing retry, fallback, and circuit breaker logic for external APIs.
- Supports OpenAI, Anthropic, Stripe, GitHub with unified response typing.
- Reduces integration complexity via consistent error handling patterns.
- Designed for ESM/CommonJS compatibility with zero runtime dependencies.

Education

Master of Computer Applications 2024 – 2026

Vivekananda Institute of Professional Studies, GGSIPU

Bachelor of Computer Applications 2021 – 2024

Bennett University